Break-Even Point and Margin of Safety

no profit, no loss!

Management Accounting

In simple words BEP is the sales level where contribution just covers fixed cost. Sales above it is the safety cushion.

1 Break-Even Point (BEP)

The level of sales (units or ₹) at which total contribution equals total fixed cost, so profit is zero.

Profit = 0 here

2 Margin of Safety (MoS)

Excess of actual (or budgeted) sales over break-even sales. It shows how far sales can fall before a loss starts.

Bigger MoS = safer firm

1. At BEP: Contribution = Fixed cost, so profit is nil
2. Above BEP, each extra unit adds its contribution to profit
3. Below BEP, the business runs at a loss
4. BEP uses contribution, not profit, in the formula



Swipe for the formula →

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THE 3 CVP FORMULAS

Break-Even Point and Margin of Safety

learn these three, score!

BEP in units

$$\text{BEP (UNITS)} = \frac{F}{\text{C per unit}}$$

F = total fixed cost; C per unit = selling price per unit – variable cost per unit

BEP in sales value

$$\text{BEP (₹)} = \frac{F}{\text{P/V ratio}}$$

P/V ratio = Contribution ÷ Sales, written as a fraction or %. Use the same ratio form in the formula.

Margin of Safety ratio

$$\text{MOS RATIO} = \frac{\text{Actual sales} - \text{BEP sales}}{\text{Actual sales}} \times 100$$

Both sales figures in the same unit, either units or ₹. Denominator is actual sales.

MOS = PROFIT ÷ P/V RATIO Margin of safety in ₹ also equals profit divided by the P/V ratio, a quick check in MCQs.

Swipe for a memory trick ⟶

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STUDY TRICK

Break-Even Point and Margin of Safety

F over C, always!

FIXED FLOATS ON CONTRIBUTION

Picture a boat (Fixed cost) floating on water (Contribution). Fixed cost always goes on top, contribution below: F over C for units, F over P/V for ₹.

HOW TO ATTEMPT IT

1. Find contribution per unit →
2. Find P/V ratio if ₹ is asked →
3. Divide fixed cost by C or P/V →
4. Subtract BEP from actual sales →
5. Divide MoS by actual sales

EXAM TIP

Check the unit asked: units or ₹. Then match the denominator, contribution per unit for units, P/V ratio for ₹.

DOUBT BUSTER

Does P/V ratio change if fixed cost rises? No. P/V ratio depends only on selling price and variable cost. A rise in fixed cost raises BEP but leaves P/V ratio unchanged.

Swipe for common mistakes →

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COMMON MISTAKES

Break-Even Point and Margin of Safety

avoid these slips!

✗ Dividing fixed cost by selling price per unit

✓ Divide fixed cost by contribution per unit (selling price – variable cost per unit).

✗ Calculating MoS % on break-even sales

✓ MoS ratio = (Actual sales – BEP sales) ÷ Actual sales × 100.

✗ Using total variable cost instead of per unit variable cost for contribution per unit

✓ Use variable cost per unit when finding contribution per unit.

✗ Mixing units and ₹ in the MoS formula

✓ Take both actual sales and BEP sales in units, or both in ₹.

TOMORROW'S CMA INTER MCQ TESTS THIS! Revise it tonight, then try the MCQ.

Save this for revision!

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